

lab04

March 25, 2026

```
[45]: from numpy import *  
      from numpy.fft import fft, ifft, fftfreq, fftshift  
      from matplotlib.pyplot import *
```

```
[46]: T = 0.15  
      t = arange(-10, 10, T)  
      rect = (abs(t) <= 0.5).astype(int)  
      rect
```

```
[46]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
          0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
          0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1,  
          1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
          0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
          0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
          0, 0])
```

```
[47]: rect.sum()  # even number of ones
```

```
[47]: 6
```

```
[50]: Frect = T*fft(rect)  
      max(abs(imag(Frect))) # not real, even though rect is real and even (symmetric).  
      ↪ Why?  
      # Because there are an even number of ones (see above)
```

```
[50]: 4.7638143839935274e-17
```

```
[51]: # How to fix it.  
      T = 0.13  
      t = arange(-10, 10, T)  
      rect = (abs(t) <= 0.5).astype(int)  
      rect.sum()  # odd number. Yeee!!
```

```
[51]: 7
```

```
[52]: Frect = T*fft(rect)  
      max(abs(imag(Frect))) # all real. Yeeee!
```

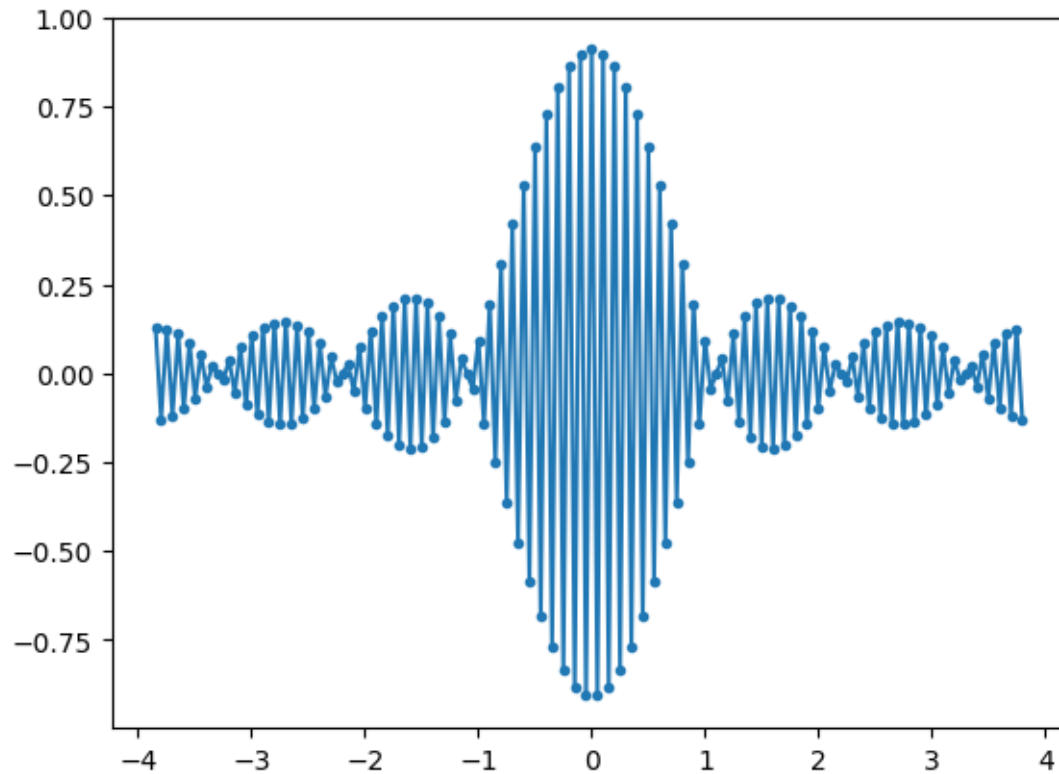
```
[52]: 4.7638143839935274e-17
```

```
[53]: f = fftfreq(len(Frect), T)
      f
```

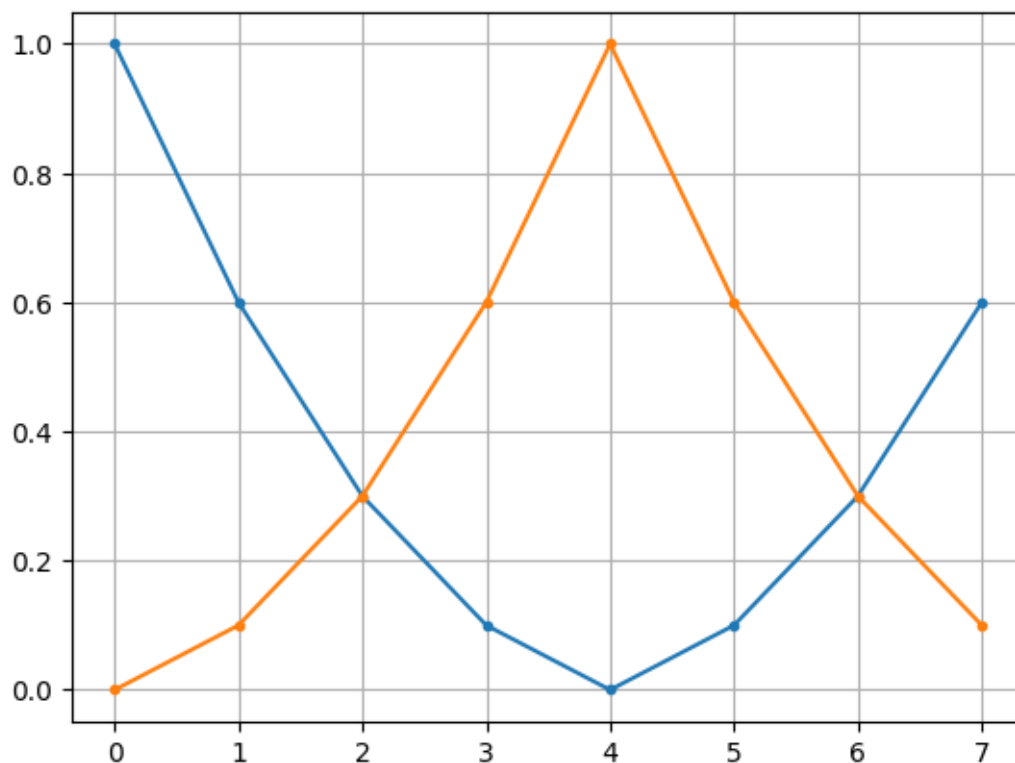
```
[53]: array([ 0.          ,  0.04995005,  0.0999001 ,  0.14985015,  0.1998002 ,
          0.24975025,  0.2997003 ,  0.34965035,  0.3996004 ,  0.44955045,
          0.4995005 ,  0.54945055,  0.5994006 ,  0.64935065,  0.6993007 ,
          0.74925075,  0.7992008 ,  0.84915085,  0.8991009 ,  0.94905095,
          0.999001 ,  1.04895105,  1.0989011 ,  1.14885115,  1.1988012 ,
          1.24875125,  1.2987013 ,  1.34865135,  1.3986014 ,  1.44855145,
          1.4985015 ,  1.54845155,  1.5984016 ,  1.64835165,  1.6983017 ,
          1.74825175,  1.7982018 ,  1.84815185,  1.8981019 ,  1.94805195,
          1.998002 ,  2.04795205,  2.0979021 ,  2.14785215,  2.1978022 ,
          2.24775225,  2.2977023 ,  2.34765235,  2.3976024 ,  2.44755245,
          2.4975025 ,  2.54745255,  2.5974026 ,  2.64735265,  2.6973027 ,
          2.74725275,  2.7972028 ,  2.84715285,  2.8971029 ,  2.94705295,
          2.997003 ,  3.04695305,  3.0969031 ,  3.14685315,  3.1968032 ,
          3.24675325,  3.2967033 ,  3.34665335,  3.3966034 ,  3.44655345,
          3.4965035 ,  3.54645355,  3.5964036 ,  3.64635365,  3.6963037 ,
          3.74625375,  3.7962038 , -3.84615385, -3.7962038 , -3.74625375,
          -3.6963037 , -3.64635365, -3.5964036 , -3.54645355, -3.4965035 ,
          -3.44655345, -3.3966034 , -3.34665335, -3.2967033 , -3.24675325,
          -3.1968032 , -3.14685315, -3.0969031 , -3.04695305, -2.997003 ,
          -2.94705295, -2.8971029 , -2.84715285, -2.7972028 , -2.74725275,
          -2.6973027 , -2.64735265, -2.5974026 , -2.54745255, -2.4975025 ,
          -2.44755245, -2.3976024 , -2.34765235, -2.2977023 , -2.24775225,
          -2.1978022 , -2.14785215, -2.0979021 , -2.04795205, -1.998002 ,
          -1.94805195, -1.8981019 , -1.84815185, -1.7982018 , -1.74825175,
          -1.6983017 , -1.64835165, -1.5984016 , -1.54845155, -1.4985015 ,
          -1.44855145, -1.3986014 , -1.34865135, -1.2987013 , -1.24875125,
          -1.1988012 , -1.14885115, -1.0989011 , -1.04895105, -0.999001 ,
          -0.94905095, -0.8991009 , -0.84915085, -0.7992008 , -0.74925075,
          -0.6993007 , -0.64935065, -0.5994006 , -0.54945055, -0.4995005 ,
          -0.44955045, -0.3996004 , -0.34965035, -0.2997003 , -0.24975025,
          -0.1998002 , -0.14985015, -0.0999001 , -0.04995005])
```

```
[54]: plot(fftshift(f), fftshift(real(Frect)), '-')
```

```
[54]: [ <matplotlib.lines.Line2D at 0x7c7aa7f613a0>]
```



```
[55]: # Question: why do we have that high 'frequency' oscillation?
# Answer: because the rectangle should be centered around the 0 index of the
#        ↳ array not in the middle
# Example:
# I want to calculate the Fourier transform of the peaked signal
# x = array([0, 0.1, 0.3, 0.6, 1.0, 0.6, 0.3, 0.1])
x = array([1, 0.6, 0.3, 0.1, 0, 0.1, 0.3, 0.6]) # the right way of representing
#        ↳ a symmetric peak to be used in FFT
plot(x, '.-') # this is how it is stored -> not symmetric
plot(fftshift(x), '.-') # this is how it is interpreted by FFT -> symmetric
grid()
```

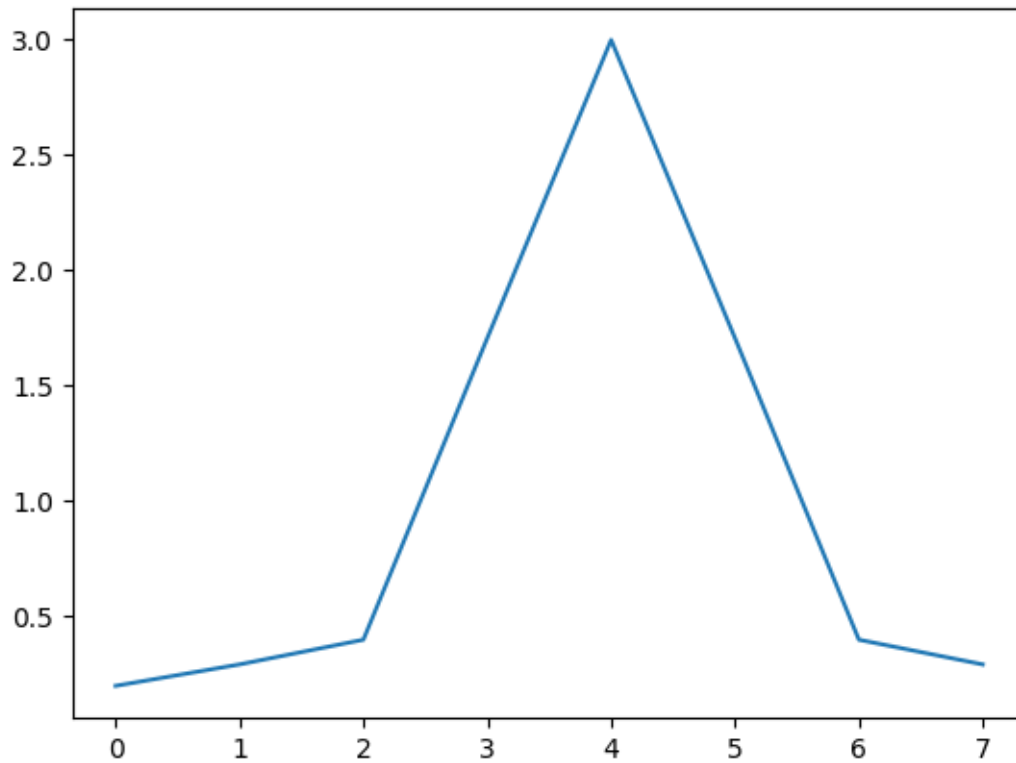


Time-shift property of the Fourier transform

$$\mathcal{F}\{x(t - t_0)\}(f) = X(f) e^{-2\pi i f t_0}$$

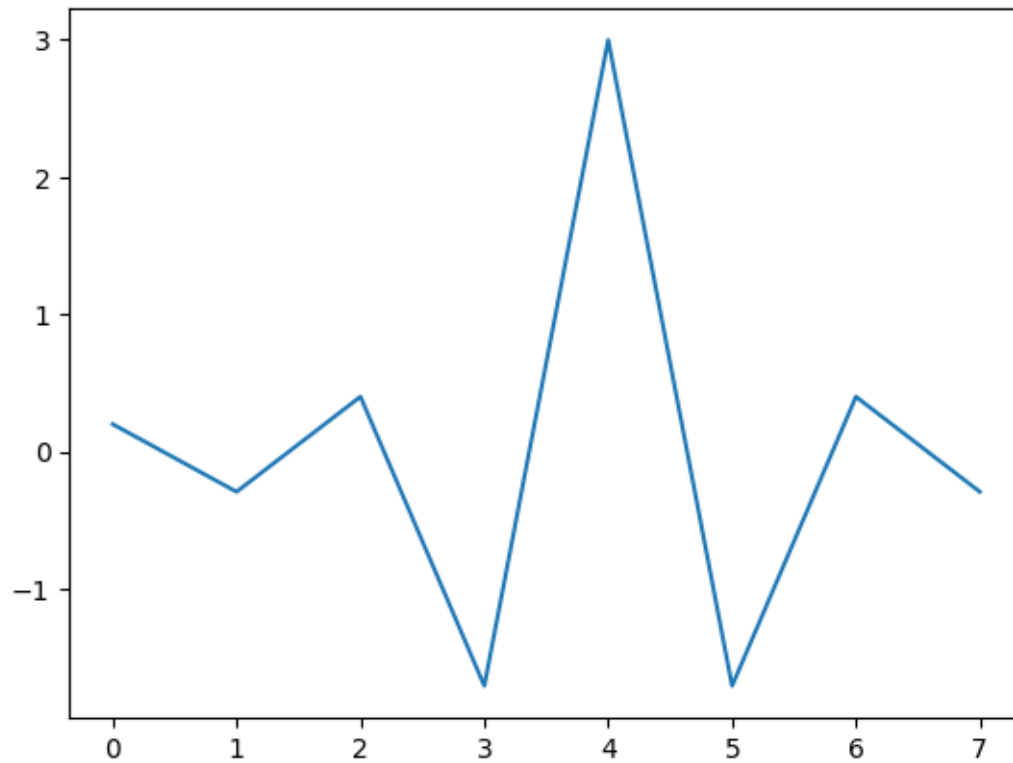
```
[56]: plot(fftfreq(N, 1.0), real(fftshift(fft(x)))) # the correct way of representing the Fourier
      ↪ transform of the peaked function
```

```
[56]: [<matplotlib.lines.Line2D at 0x7c7aa7e46720>]
```



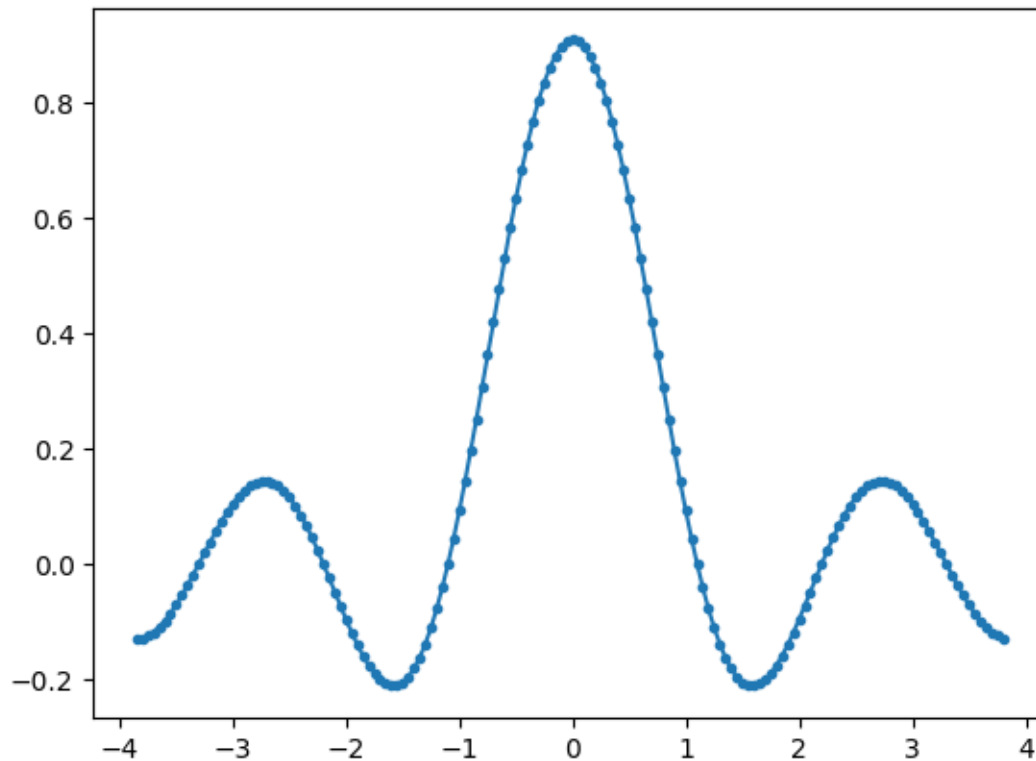
```
[57]: plot(fftshift(real(fft(fftshift(x))))) # we have an extra oscillation because  
      ↪ of the fftshift(x). The outer fftshift is just for the better visualization  
      ↪ of the FFT result.
```

```
[57]: [<matplotlib.lines.Line2D at 0x7c7aa7ec4590>]
```



```
[58]: #getting back to the 'rect'-s FFT  
#let's shift the function so that it is centered on the 0 index  
Frect = T*fft(fftshift(rect))  
plot(fftshift(f), fftshift(Frect),'.-') # Yeeee!
```

```
[58]: [<matplotlib.lines.Line2D at 0x7c7aa7d44560>]
```

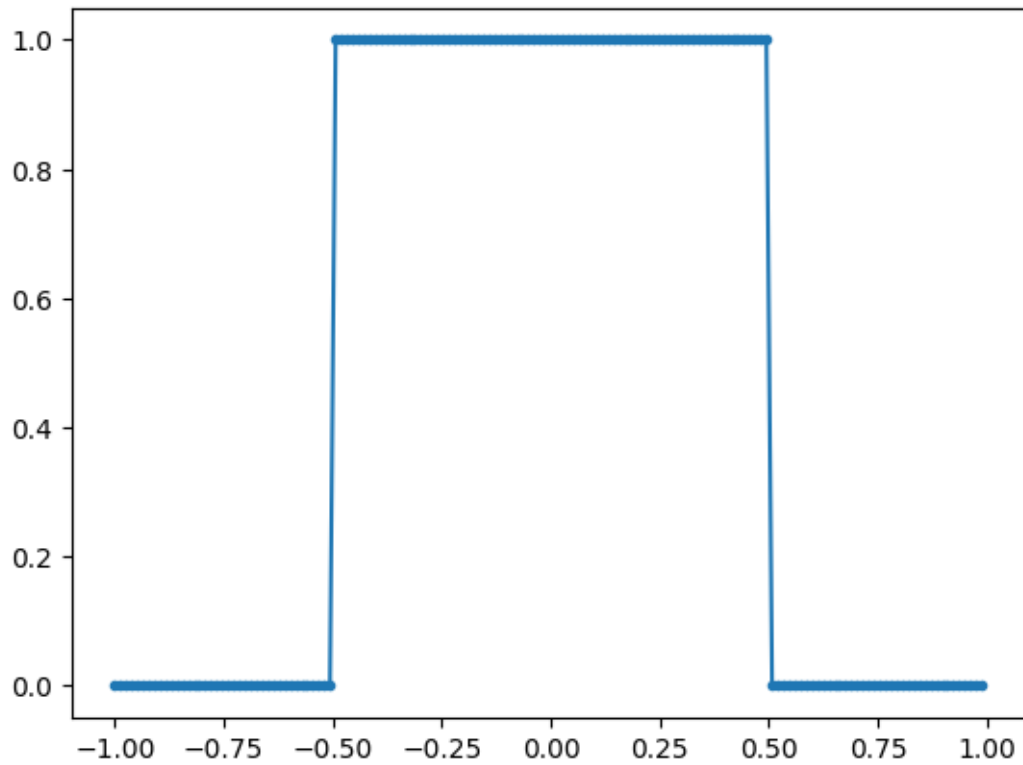


```
[59]: # Question: Why just three maxima?
# Answer: because of the 'thin' rectangle. Rule: small temporal scale -> large
      ↪ frequency scale
# Question: How to make it have more waves.
# Answer: increase temporal scale (without changing the size of the array)
T = 0.013 # higher resolution
t = arange(-1, 1, T) # smaller interval (with respect to the rectangle's width)
rect = (abs(t) <= 0.5).astype(int)
rect.sum() # odd number. Yeee!!
```

```
[59]: 77
```

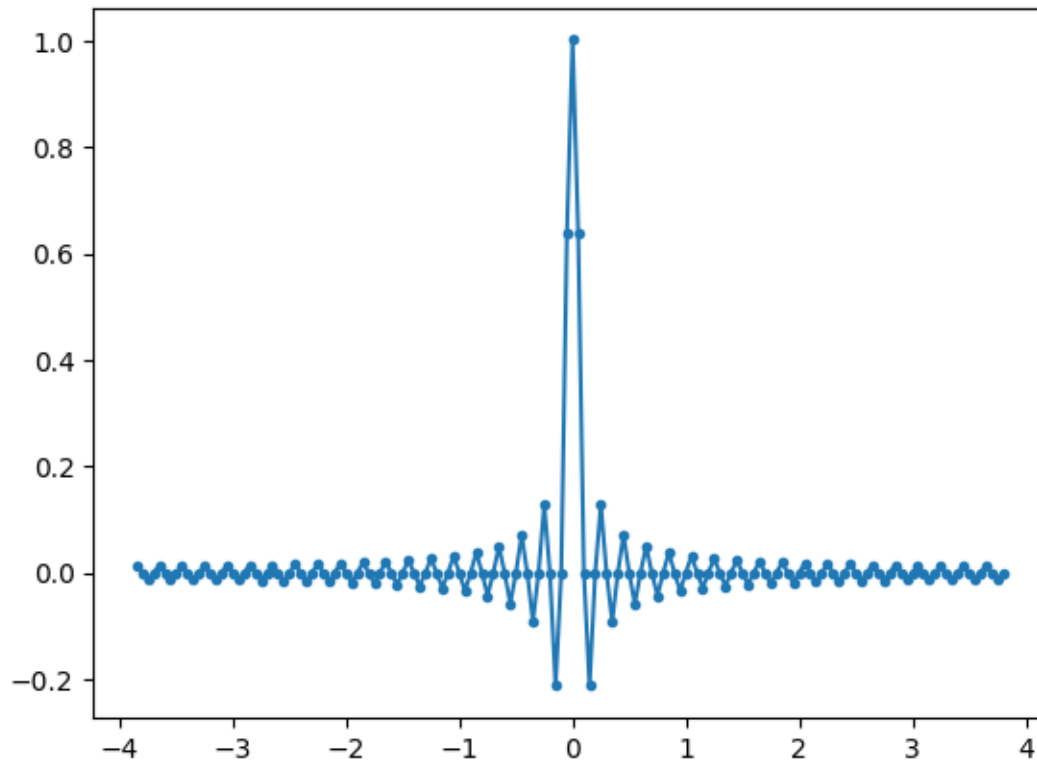
```
[60]: plot(t, rect, '.-')
```

```
[60]: [<matplotlib.lines.Line2D at 0x7c7aa7d9a4b0>]
```



```
[61]: Frect = T*fft(fftshift(rect))  
      plot(fftshift(f), fftshift(Frect), '.-') # Yeee! large temporal scale -> small ␣  
      ↪ frequency scale
```

```
[61]: [<matplotlib.lines.Line2D at 0x7c7aa7c30170>]
```

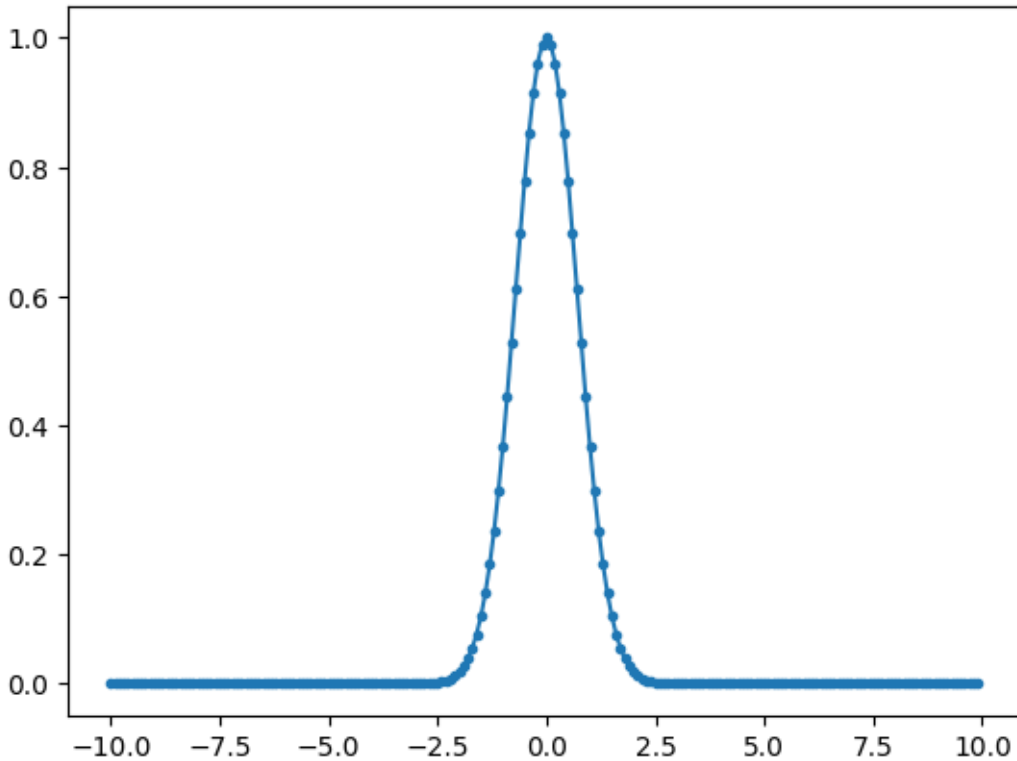



Let's check this:

$$\mathcal{F} \left[e^{-t^2} \right] = \sqrt{\pi} e^{-\pi^2 f^2}$$

```
[62]: T = 0.1
      t = arange(-10, 10, T)
      x = exp(-t**2)
      plot(t, x, '.-')
```

```
[62]: [<matplotlib.lines.Line2D at 0x7c7aa7c5f260>]
```



```
[63]: fftshift(x) # check whether the peak is indeed at index 0. Yeee!
```

```
[63]: array([1.00000000e+00, 9.90049834e-01, 9.60789439e-01, 9.13931185e-01,
 8.52143789e-01, 7.78800783e-01, 6.97676326e-01, 6.12626394e-01,
 5.27292424e-01, 4.44858066e-01, 3.67879441e-01, 2.98197279e-01,
 2.36927759e-01, 1.84519524e-01, 1.40858421e-01, 1.05399225e-01,
 7.73047404e-02, 5.55762126e-02, 3.91638951e-02, 2.70518469e-02,
 1.83156389e-02, 1.21551783e-02, 7.90705405e-03, 5.04176026e-03,
 3.15111160e-03, 1.93045414e-03, 1.15922917e-03, 6.82328053e-04,
 3.93669041e-04, 2.22629857e-04, 1.23409804e-04, 6.70548243e-05,
 3.57128496e-05, 1.86437423e-05, 9.54016287e-06, 4.78511739e-06,
 2.35257520e-06, 1.13372714e-06, 5.35534780e-07, 2.47959602e-07,
 1.12535175e-07, 5.00621802e-08, 2.18295780e-08, 9.33028757e-09,
 3.90893843e-09, 1.60522806e-09, 6.46143177e-10, 2.54938188e-10,
 9.85950558e-11, 3.73757133e-11, 1.38879439e-11, 5.05825274e-12,
 1.80583144e-12, 6.31928589e-13, 2.16756888e-13, 7.28772410e-14,
 2.40173478e-14, 7.75840208e-15, 2.45659537e-15, 7.62445991e-16,
 2.31952283e-16, 6.91675398e-17, 2.02171585e-17, 5.79231289e-18,
 1.62666462e-18, 4.47773244e-19, 1.20818202e-19, 3.19536672e-20,
 8.28367701e-21, 2.10493998e-21, 5.24288566e-22, 1.28001532e-22,
 3.06319086e-23, 7.18533564e-24, 1.65209178e-24, 3.72336312e-25,
 8.22528065e-26, 1.78106663e-26, 3.78027784e-27, 7.86468594e-28,
```

```

1.60381089e-28, 3.20581932e-29, 6.28114815e-30, 1.20629393e-30,
2.27081292e-31, 4.19009319e-32, 7.57844527e-33, 1.34354020e-33,
2.33472278e-34, 3.97680310e-35, 6.63967720e-36, 1.08661064e-36,
1.74307090e-37, 2.74075528e-38, 4.22415241e-39, 6.38150345e-40,
9.44975498e-41, 1.37161491e-41, 1.95145238e-42, 2.72143414e-43,
3.72007598e-44, 2.72143414e-43, 1.95145238e-42, 1.37161491e-41,
9.44975498e-41, 6.38150345e-40, 4.22415241e-39, 2.74075528e-38,
1.74307090e-37, 1.08661064e-36, 6.63967720e-36, 3.97680310e-35,
2.33472278e-34, 1.34354020e-33, 7.57844527e-33, 4.19009319e-32,
2.27081292e-31, 1.20629393e-30, 6.28114815e-30, 3.20581932e-29,
1.60381089e-28, 7.86468594e-28, 3.78027784e-27, 1.78106663e-26,
8.22528065e-26, 3.72336312e-25, 1.65209178e-24, 7.18533564e-24,
3.06319086e-23, 1.28001532e-22, 5.24288566e-22, 2.10493998e-21,
8.28367701e-21, 3.19536672e-20, 1.20818202e-19, 4.47773244e-19,
1.62666462e-18, 5.79231289e-18, 2.02171585e-17, 6.91675398e-17,
2.31952283e-16, 7.62445991e-16, 2.45659537e-15, 7.75840208e-15,
2.40173478e-14, 7.28772410e-14, 2.16756888e-13, 6.31928589e-13,
1.80583144e-12, 5.05825274e-12, 1.38879439e-11, 3.73757133e-11,
9.85950558e-11, 2.54938188e-10, 6.46143177e-10, 1.60522806e-09,
3.90893843e-09, 9.33028757e-09, 2.18295780e-08, 5.00621802e-08,
1.12535175e-07, 2.47959602e-07, 5.35534780e-07, 1.13372714e-06,
2.35257520e-06, 4.78511739e-06, 9.54016287e-06, 1.86437423e-05,
3.57128496e-05, 6.70548243e-05, 1.23409804e-04, 2.22629857e-04,
3.93669041e-04, 6.82328053e-04, 1.15922917e-03, 1.93045414e-03,
3.15111160e-03, 5.04176026e-03, 7.90705405e-03, 1.21551783e-02,
1.83156389e-02, 2.70518469e-02, 3.91638951e-02, 5.55762126e-02,
7.73047404e-02, 1.05399225e-01, 1.40858421e-01, 1.84519524e-01,
2.36927759e-01, 2.98197279e-01, 3.67879441e-01, 4.44858066e-01,
5.27292424e-01, 6.12626394e-01, 6.97676326e-01, 7.78800783e-01,
8.52143789e-01, 9.13931185e-01, 9.60789439e-01, 9.90049834e-01])

```

```

[64]: X = T*fft(fftshift(x))
      f = fftfreq(len(x), T)
      # check if its real
      max(abs(imag(X))) # Yeee!

```

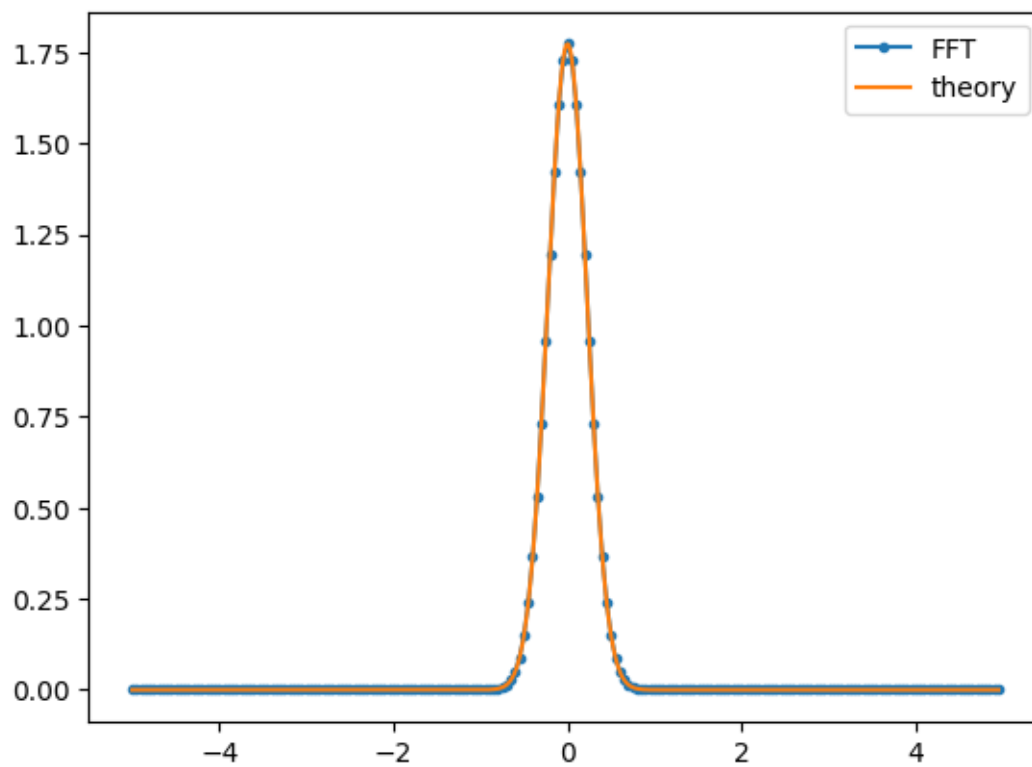
[64]: 5.338483201215879e-14

```

[65]: plot(fftshift(f), fftshift(X), '.-', label='FFT')
      # compare with theory
      ff = linspace(f.min(), f.max(), 10000)
      plot(ff, sqrt(pi)*exp(-pi*pi*ff**2), label='theory')
      legend() # Yeeee!

```

[65]: <matplotlib.legend.Legend at 0x7c7aa7d71880>



[]:

[]: